



TECHNICAL MANUAL FOR
USE AND MAINTENANCE

LIFETEK XP-R 33/55/100

ENVIRONMENTAL SAMPLER

REV. 02 – 09/2016

FILE: LIFETEK 33-55-100_3.6_REV.02_ENGLISH

MEGA SYSTEM s.r.l.

1 INTRODUCTION

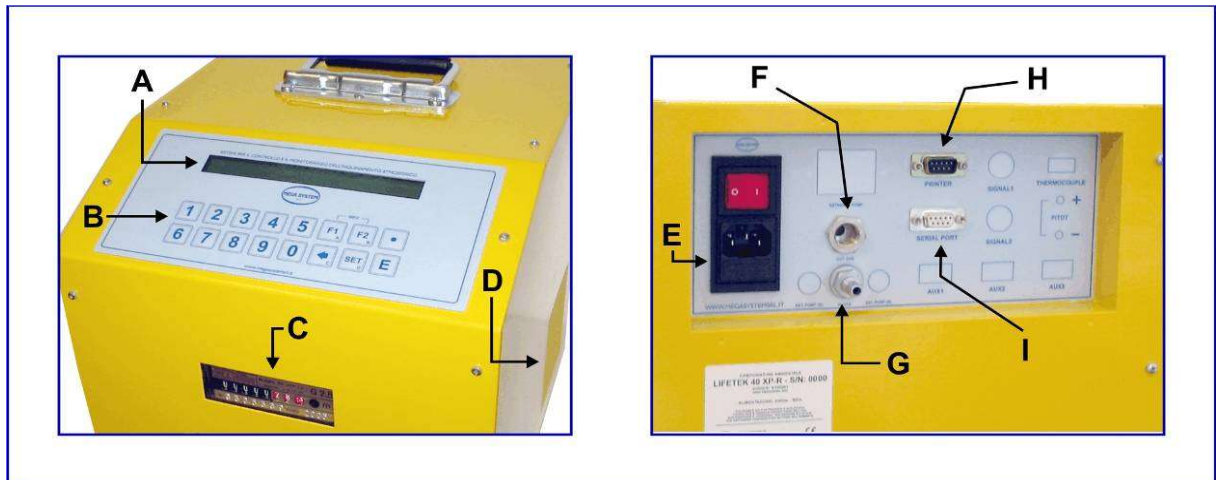
The personal sampler of the LIFE series has been designed to suitable for personal sampling according the European law in terms of health protection and worker's safety while working (D.L. 25 of 02/02/2002 – D.L. 626/94 – D.P.R. And further modifications).

They can also be used to do total dust, thoracic fraction ad respirable (PM10 and PM2,5 –US-EPA), asbestos fibers, of gas and vapors and isokinetics sampling in EMISSION.



The pump type of the sampler is a rotary vane pump for LIFETEK 100, membrane pump for LIFETEK 33 / 55. The pump provides high precision in the fluid regulation over all its functionality range. The **high prevalence** of the pump unite with the **electronic control** and management system can automatically compensate the variable load loss caused by the filter clogging and keep the range constant over all the sampling time.

2 DESCRIPTION OF THE INSTRUMENT



- A** Alphanumeric LCD display (40x2)
- B** Multifunction keyboard
- C** Not resettable Dry volumetric counter
- D** Air intake for pump ventilation
- E** Power supply connector (230Vac – 50Hz) and ON/OFF switch
- F** Air dump
- G** Quick sampling line connector (Intake)
- H** Serial port for printer connection
- I** Serial port for PC data transfert via cable – ISOCHECK TSB Connection

On the LIFETEK 33/55/100 XP-R the AUX1 port is used to connect an ISOCHECK using a serial standard cable (optional)

IMPORTANT!

Difference between 33/55/100:

LIFETEK 33:

- **Working range: 0.2 ÷ 25 lpm**
- **On the back panel of the instrument there are a red (+) and a black connector (-) that can be used to connect a 12Vcc battery (optional) to supply the instrument and do sampling without an electrical grid.**
- **To use the instrument with the external battery is INDISPENSABLE to activate the ON/OFF switch.**

LIFETEK 55:



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- **Working range: 0.25 ÷ 40 lpm**
- **On the back panel of the instrument there are a red (+) and a black connector (-) that can be used to connect a 24Vcc battery (optional) to supply the instrument and do sampling without an electrical grid.**
- **To use the instrument with the external battery is INDISPENSABLE to activate the ON/OFF switch.**

LIFETEK 100:

- **Working range: 5 ÷ 70 lpm**
- **Battery use is not supported**

3 SOFTWARE DESCRIPTION

3.1 DATA INSERTION METHODS

The software has been created to simplify the menu' navigation and the data settings. It'll follow a description of some simple operations for an option selection or a proposed data modification.

Option selection

Press the proposed button from the menu.

Proposed value change

Press the SET key to set the new value and confirm it with the E button

3.2 DATE / TIME SETTINGS AND MAIN MENU'

Place the ON-OFF switch on ON position to power-on the instrument. It will appear the following mask:

```
***** MEGA SYSTEM S.R.L *** MILANO *****
*LIFETEK 55XP-R R3.5 2010 S/N 0000 *
```

... after a few seconds ...

```
FREE MEMORY =100%
01/01/06 - 12:30:00 <E>CONT <S>SET TIME
```

Press the E button to get direct access to the MAIN MENU or press SET to change the current date and time. The first proposed parameter to change is DAY. Change the data and press E to confirm the modification. Go on with the same procedure to change the related month, year, hour and minute data.

```
SETTING DATE / TIME
SET DAY (1-31) 01
```



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```
SETTING DATE / TIME
SET MONTH      (1-12)      01
```

```
SETTING DATE / TIME
SET YEAR       (06-XX)     06
```

```
SETTING DATE / TIME
SET HOURS      (0-23)     12
```

```
SETTING DATE / TIME
SET MINUTES    (0-59)     30
```

At the end you will return to the following mask:

```
SET DATE / HOUR
01/01/06 - 12:30:00  <E>CONT <S>SET TIME
```

Press E to access the MAIN MENU.

Use the F1 and F2 keys to scroll the main menu'. Press the E key to select the desired program. Hereunder are reported the compatible programs.

```
**SELECT PROGRAM**  F1>> F2<<  <E>CONT
TEST (TEMPERATURE-LEAK  -LOAD  -FLOW  )
```

1^ mask of the main menu'

```
**SELECT PROGRAM**  F1>> F2<<  <E>CONT
ENVIRONMENTAL DUST SAMPLING
```

```
**SELECT PROGRAM**  F1>> F2<<  <E>CONT
DUCT DUST SAMPLING
```

```
**SELECT PROGRAM**  F1>> F2<<  <E>CONT
ISOKINETIC SAMPLING WITH ISOCHECK
```

```
**SELECT PROGRAM**  F1>> F2<<  <E>CONT
ISOKINETIC SAMPLING WITH TSB
```

```
**SELECT PROGRAM**  F1>> F2<<  <E>CONT
MEMORY MANAGER
```

```
**SCELTA PROGRAMMA** F1>> F2<<  <E>CONT
PM10 HEAD SAMPLING
```



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```
**SCELTA PROGRAMMA**  F1>> F2<<  <E>CONT
      SELECT_8 ASSISTANCE
```

This last two menu are active only in the XP-R-PM version

Important note:

- During the first use of the instrument is suggested to clean the memory
- Before you begin the sampling is suggested to check the setup parameters
- Before you begin the isokinetic sampling with the ISO CHECK we recommended to do the TEST – *Charging Table* with the sampling line connected to the instrument, so to optimize the flow adjustment during the sampling (check the TEST paragraph).

3.3 TEST (TEMPERATURE-VACUUM-LOAD-FLOW)

Use the F1 and F2 keys to scroll the main menu'. Select "TEST (TEMPERATURE-VACUUM-LOAD-FLOW)" and press the E button to gain access to the program.

```
0=TEST TEMP  1=LEAK TEST  2=TAB.LOAD.
3=FLOW      4=EXIT        SELECT? =4
```

In the TEST program is possible to do the following attempt:

```
OPTION 0      TEMPERATURE TEST
OPTION 1      VACUUM TEST
OPTION 2      LOAD TABLE
OPTION 3      FLOW TEST
OPTION 4      EXIT – GO BACK TO THE FIRST MASK OF THE MAIN MANU'
```

Confirm the selected option with the E button to go on.

3.3.1 TEMPERATURE TEST

Select this option to test the proper functioning of the temperature sensor situated in the volumetric counter.

```
TEST TEMPERATURE      <←> EXIT
TEMP=20.6             mv=2058
```

Press the key ← (C) to end and go back to the program menu' "TEST (TEMPERATURE-VACUUM - LOAD-FLOW)".

3.3.2 VACUUM TEST

Select this option to test the capacity of the sampling line.



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To do the capacity test is necessary to fully connect the sampling line to its links and close the line itself in the intake point.

```
LEAK TEST      <E> START
BALANCED VACUUM=1  CONSTANT VACUUM=2
```

Press 1 to proceed with the balancing of the vacuum during the 60 seconds of the test
Press 2 to proceed with the balancing of the vacuum at the end of the 60 seconds of the test

```
BALANCED VACUUM TEST  <C> OUT
CLOSE THE SAMPLING LINE <E> CONT.
```

Or

```
CONSTANT VACUUM TEST  <C> OUT
CLOSE THE SAMPLING LINE <E> CONT.
```

Close the sampling line and press E to continue.

```
LEAK TEST %
VACUUM mmHg = 400
```

Set the selected Vacuum value. This value is the one desired during the sample.

```
WAIT VACUUM REGULATION
VACUUM = 350      PWM = 2000
```

The pump starts to reach the desired vacuum value. Until the pump reach the value and:

```
TEST IN PROGRESS  <C> OUT
SECONDS=30  LITRES=0.00  VACUUM=400
```

The test lasts 60 seconds within the software count the potential loss liters

IN CASE OF LOSS: if has been selected the balanced vacuum (N=1), the pump will be turned on to maintain constant the vacuum. Otherwise, the pump will start to works at the end of the test to reach selected

At the end:

```
TEST FINISHED
SAMPLING FLOW EXPECTED = 15
```



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Set the sampling flow expected for the sample to be able to calculate the error percentage of loss

```
CALCULATED LOSS =0.00% 1t=0.0
<E> CONT.
```

If the percentage exceed what expected, (see the norm), verify the line and execute again the test

Press the key ← (C) to end earlier the test and go back to the menu' of the "TEST (TEMPERATURE- VACUUM – LOAD-FLOW)".

3.3.3 LOAD TABLE

Select this option to store the load table.

Before starting the isokinetic sampling with the ISOCHECK is recommended to do the TEST Load Table with the charging line connected to the instrument. In this way the flow adjustment is optimized during the sampling.

To do the test is necessary to properly make all the connection to the sampling line.

WARNING!!! - Don't do this test with the in line glass impingers.

```
LOAD TABLE PUMP – CONNECT THE LINE
<E> START <←> EXIT
```

Press the key ← (C) to end and go back to the "TEST (TEMPERATURE- VACUUM – LOAD-FLOW)" program menu'.

Press the E button to start the Load Table memorization test.

```
LOAD TABLE PUMP – CONNECT THE LINE
X.XX YYY ind=Z
```

At the beginning of the test the pump will start and the speed will be increased using default step. On the display will be showed the result of the parameter XXX and YYY for the table creation.

The test last about 5 minutes.

At the end of this test the device will automatically go to the "TEST (TEMPERATURE- VACUUM – LOAD-FLOW)" program menu'.



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3.3.4 FLOW TEST

Using this option is possible to do an intake flow check, with a determinate pneumatic load, setting the percentage of the pump power.

```
FLOW TEST (LENGHT 60 sec) <C>=STOP
SET PERCENTAGE (1-100) = 100
```

To test the maximum flow with a determinate pneumatic load set the value to 100.
To test the minimum flow with a determinate pneumatic load set the value between 10 and 20.

Press the key ← (C) to change the percentage value proposed and set the choose one.

Press E button to start the test and initiate the pump.

```
FLOW TEST (LENGHT 60 sec) <C>=STOP
*** WAIT! FLOW REGULATION ***
```

... after a few seconds the count will start...

```
FLOW TEST (LENGHT 60 sec) <C>=STOP
TEST IN PROGRESS sec.=XX
```

... at the ending...

```
FLOW TEST (LENGHT 60 sec) <C>=STOP
<←> EXIT FLOW = 49.7 l/min
```

At the end of the test the pump will stop and will be displayed the aspiration flow in litres for minutes.

Press the key ← (C) to end and go back to the "TEST (TEMPERATURE- VACUUM – LOAD-FLOW)" program menu'.

3.3.5 EXIT - GO BACK TO THE FIRST MASK OF THE MAIN MENU'

Select this option to go back to the 1^ mask of the main menu'.

```
**SELECT PROGRAM** F1>> F2<< <E>CONT
TEST (TEMPERATURE-LEAK -LOAD -FLOW )
```

1^ mask of the main menu'



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3.4 ENVIRONMENTAL DUST SAMPLING

WARNING: At the start up the instrument check the available memory and give the following message if it is lower than 25%: "MEMORY ALMOST FULL". In this case press the F1 button to continue and proceed to the download of the saved data on a PC. After that, before starting a new sampling, is better to execute the memory erase.

Use the F1 and F2 keys to scroll the main menu'. Select "ENVIRONMENTAL DUST SAMPLING" and press the E button to gain access to the program.

*****	ENVIRONMENT SAMPLING	*****
1=DATA	2=START	3=EXIT
		SCELTA? =3

In the program is possible to select one of the following option:

OPTION 1	DATA
OPTION 2	START
OPTION 3	EXIT – GO BACK TO THE FIRST MASK OF THE MAIN MENU'

Confirm the selected option with the E button to go on.

3.4.1 DATA

Before starting the sampling is possible to do a check/edit of the programming data of the sampling itself. The programming data will be stored in memory until the next modification.

ENVIRO SAMPLING	<S>SET	<E>CONT
ASPIRATION FLOW (0.25-40) l/min=10.00		

Set the intake flow (range 5-70 for the LIFETEK 100).

Press the E button to go on or press the SET button to change the proposed value.

ENVIROMENT SAMPLING	<S>SET	<E>CONT
1=SET TIME	2=SET VOLUME	=1

Set the programming methods. Using the first option will be set the sampling time, otherwise, with the option 2, will be set the sampler volume.

Press E to go on or press the SET button to change the proposed value

ENVIROMENT SAMPLING	<S>SET	<E>CONT
ELAPSED TIME (2-9999)	min=	60

The sampling length settings mask will be showed if previously was choose the first option, instead will be showed the following mask:



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ENVIROMENT SAMPLING	<S>SET	<E>CONT
LITERS to SAMPLE (1-65000)	l=	100

Press E to go on or press the SET button to change the proposed value.

ENVIROMENT SAMPLING	<S>SET	<E>CONT
STOP FLOW SAMPLING (0-xx)	l/min=	0.10

Set the stop flow sample.

The stop flow sample is the flow under which the sampling ends for a filter clogging (excessive load) cause.

Press E to go on or press the SET button to change the proposed value.

ENVIROMENT SAMPLING	<S>SET	<E>CONT
SAMPLING DELAYED START (1-9999)min=		0

Set the sampling delay start.

Press E to go on or press the SET button to change the proposed value.

ENVIROMENT SAMPLING	<S>SET	<E>CONT
NORMALIZED TEMPERATURE (0-50)	C=	20

Set the normalization temperature.

Press E to go on or press the SET button to change the proposed value.

*****	ENVIROMENT SAMPLING	*****
1=DATA	2=START	3=EXIT
		SELECT? =3

At the end it will go back to the "ENVIRONMENTAL DUST SAMPLING" program menu'.

3.4.2 START

After you have done the pneumatic link is possible to select this option to start the sampling.

The sampling will be done using the settings in the "DATA" section.

*****	ENVIROMENT SAMPLING	*****
***	WAIT! FLOW REGULATION	***

... after a few seconds ...

TIME	T.cont	VOLUME	FLOW	vacuum
02.0	20.5	00020	9.96	098

During the sampling will be displayed the following data:



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TIME	Time (minutes) – This value increase while the sampling goes on
T.cont	Temperature (°C) at the gas meter-This value can change while the sampling goes on
VOLUME	Sampled Volume (litres) - This value can change while the sampling goes on
FLOW	Effective Flow (l/min)
VACUUM	Vacuum of sampling line

During the sampling is available the button ← (C). Pressing this button is possible to interrupt earlier the sample. In the final data will be reported that the sampling was interrupted by the operator.

At the end of the sampling will be displayed the following mask:

END SAMPLING DATA MANAGER

At the end of the sampling the pump will stop and the data stored and the instrument will automatically display the first mask of the main menu'.

The memorized data can be displayed, printed, downloaded to a PC or deleted using the memory settings menu' (see the relative paragraph).

EXIT - GO BACK TO THE FIRS MASK OF THE MAIN MENU'

Select this option to go back to the 1^ mask of the main menu'.

SELECT PROGRAM F1>> F2<< <E>CONT TEST (TEMPERATURE-LEAK-LOAD-FLOW)

1^ mask of the main menu'



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3.5 DUST DUCT SAMPLING

WARNING: At the start up the instrument check the available memory and give the following message if it is lower than 25%: "MEMORY ALMOST FULL". In this case press the F1 button to continue and proceed to the download of the saved data on a PC. After that, before starting a new sampling, is better to execute the memory erase.

Use the F1 and F2 keys to scroll the main menu'. Select "DUST DUCT SAMPLING" and press the E button to gain access to the program.

DUST DUCT SAMPLING			
1=DATA	2=START	3=EXIT	SELECT? =3

In the program is possible to select one of the following option:

OPTION 1	DATA
OPTION 2	START
OPTION 3	EXIT – GO BACK TO THE FIRST MASK OF THE MAIN MENU'

Confirm with E to go on with the selected option.

3.5.1 DATA

Before starting the sampling is possible to do a check/edit of the programming data of the sampling itself. The programming data will be stored in memory until the next change.

STACK SAMPLING	<S>SET	<E>CONT
1=SET TIME	2=SET VOLUME	=1

Set the programming mode. Using the first option will be set the sampling time, otherwise, with the option 2, will be set the volume sampling.

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
ELAPSED TIME (2-9999)		min= 60

The settings mask of the sampling length will be displayed if at the previously question the answer was the option 1, otherwise will be showed the following mask:

STACK SAMPLING	<S>SET	<E>CONT
SAMPLING LITERS (1-65000)		1= 100

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
STACK SHAPE	0=CIRC 1=RECT	= 1



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Set the duct shape.

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
STACK DIAMETER	cm	= 100

... option displayed if the duct shape = circular ...

STACK SAMPLING	<S>SET	<E>CONT
GREATER SIDE	cm	= 100

... option displayed if the duct shape = rectangular ...

STACK SAMPLING	<S>SET	<E>CONT
MINOR SIDE	cm	= 90

... option displayed if the duct shape = rectangular ...

Set the dimensional characteristic of the chimney. Depending of the chimney shape previously selected, will be asked to set the **measure in cm of the diameter** or the sides.

For the rectangular chimney is possible to insert the same values.

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
CONSTANT K OF THE MEASURE TUBE		= 0.83

Set the constant K of the measure duct. The constant will be used in the calculation as the UNI 10 169 (out of root).

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
DRY GAS DENSITY	(0 C.)	kg/m3 = 1.293

Set the dry gas density (0°C – 1013 mBar).

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
BAROMETRIC PRESSURE	hpa	= 1013

Set the barometric pressure.

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
MEASURE UNIT (0=mmH2O 1=PA)		= 0

Set the measure unit.

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
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STACK STATIC PRESSURE mmH2O =	30
-------------------------------	----

Set the static duct pressure measured as the UNI 10169(DPe).

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
DIFFERENTIAL PRESSURE	mmH2O =	10

Set the differential pressure.

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
WALL EFFECT (WAF)	=	0.995

Set wall factor (WAF) as requested by UNI EN ISO 16911-1.

Press E to go on or press the SET button to change the proposed value

STACK SAMPLING	<S>SET	<E>CONT
MOISTURE	% =	10

Set the condensation %.

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
NORM. TEMPERATURE (0-50)	C=	20

Set the normalization temperature.

Press E to go on or press the SET button to change the proposed value.

STACK SAMPLING	<S>SET	<E>CONT
STACK TEMPERATURE	C. =	1000

Set the duct temperature.

Press E to go on or press the SET button to change the proposed value.

NOZ	2	4	6	8	10	<E>CONT
	0.9	3.6	8.2	14.5	22.7	

At this point will be proposed a table containing a series of nozzles and their isokinetic flow.

Press E to go on.

STACK SAMPLING	<S>SET	<E>CONT
NOZZLE DIAMETER	mm =	6

At this point will be proposed a table containing a series of nozzles and their isokinetic flow.



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Press E to go on

```
NOZZLE DIAMETER = 6    <SET>CHANGE <E> CONT
ISOKINETIC FLOW =8.16
```

Press E to go on or press the SET button to go back to the previous mask and change the nozzle's diameter and show again their isokinetic flows.

```
DUCT DUST SAMPLING
1=DATAI 2=START 3=EXIT      SELECT? =3
```

At the ending the instrument will go back to the "DUST DUCT SAMPLING" program menu'.

3.5.2 START

After you have done the pneumatic link is possible to select this option to start the sampling.

The sampling will be done according the settings in the DATA option.

```
*****   STACK SAMPLING   *****
*** WAIT!  FLOW REGULATION ***
```

... after a few seconds ...

TIME	T.mtr	VOLUME	ISO.FL	FLOW	%ERR
02.0	20.5	00020	8.15	8.14	-0.1

... 1^ mask (displayed alternately at the 2^) ...

DP mmH2O	T.mtr	SP.stack	VELOCITY	NZ
10.0	31,1	030	22.47	06

... 2^ mask (displayed alternately at the 1^) ..

VEL m/sec	SCOPE Nm3/h	VOID mmhg
8.43	000223	0001

... 3^ mask (displayed alternately at the 1^) ..

During the sampling will be displayed the following data:

1^ MASK:

TIME Time (minutes) - This value increase while the sampling goes on



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TEMP	Temperature (°C) at the counter - This value change while the sampling goes on
VOLUME	Sampled Volume (litres) - This value increase while the sampling goes on
ISO.FL	ISOKINETIC Flow (l/min)
FLOW	EFFECTIVE Flow (l/min)
%ERR	Error (%) of the effective flow in comparison of the set one

2^ MASK:

DP	Differential pressure (value received by ISOCHECK)-UM.: mmH2O or Pa
T.cond	Duct temperature (value received by ISOCHECK)
SP.duct	Static duct pressure (value received by ISOCHECK)-UM.:mmH2O or Pa
VELOCITY	Velocity
NZ	Set nozzle

3^ MASK:

VEL	Velocity
FLOW RATE	(Nm3/h) – stack emission flow rate hour average
VOID	Residual Vacuum (mmhg) of sampling line

During the sampling is active the key ← (C). Pres this button to be able to change the values of the thermodynamic parameters or stop the sampling earlier.

At the end of the sample will be displayed the following mask:

END SAMPLING DATA MANAGER

At the end of the sample the pump will stop, the sampling results are stored and the device will automatically go back to the first mask of the main menu'.
The stored data can be displayed, printed, downloaded to a PC or deleted using the data management menu' (see the relative paragraph).

EXIT - GO BACK TO THE FIRST MASK OF THE MAIN MENU'

Select this option to go back to the 1^ mask of the main menu'.

SCELTA PROGRAMMA F1>> F2<< <E>CONT TEST (TEMPERATURA-TENUTA-CARICO-FLUSSO)

1^ mask of the main menu'



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3.6 ISOKINETIC SAMPLING WITH ISOCHECK

WARNING: At the start up the instrument check the available memory and give the following message if it is lower than 25%: "MEMORY ALMOST FULL". In this case press the F1 button to continue and proceed to the download of the saved data on a PC. After that, before starting a new sampling, is better to execute the memory erase.

With the "ISOKINETIC SAMPLING WITH ISOCHECK" program is possible to do AUTOMATIC ISOKINETIC sampling.

To do an automatic isokinetic sample is necessary to have an automatic processor ISOCHECK type XP-SR, since that also the processor must have the REMOTE LINK interface.

- Before starting is necessary to have done a series of measures with the processor only, so is possible to choose the appropriate nozzle for the sampling.
- After that the pneumatic and electric link must have been connected to the sampler.
- Do the link between the sampler and the processor using the serial cable and the adapter...
- Set up the processor at the measure for the automatic isokinetic sample (REMOTE LINK) using the relative technical manual.

On the LIFETEK sampler proceed as described:

Use the F1 and F2 keys to scroll the main menu'. Select "ISOKINETIC SAMPLING WITH ISOCHECK"

and press the E button to gain access to the program.

```
***  AUTOMATIC ISOKINETIC SAMPLING  ***
ISOCHECK CONNECTION TEST
```

As soon the program shows up, it will be done a link test to the processor. If the processor was correctly linked and set up for the REMOTE LINK program, it will transfer the SETUP from the processor to the sampler. The SETUP contain all the data relative of the chimney, density, condensate and K constant.

In the case that the ISOCHECK LINK TEST will be negative, it will be showed the failure of the link and will be impossible to do the automated isokinetic sampling.

In this case do a check of the electric links, an integrity check of the cable and the adaptor and a check of the programmed processor. In the case the problem will not be solved contact the technical support.

```
***  AUTOMATIC ISOKINETIC SAMPLING  ***
DOWLOADING SETUP FROM  ISOCHECK DONE
```

... after a few seconds ...



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```
***  AUTOMATIC ISOKINETIC SAMPLING ***
1=DATA  2=START  3=EXIT      SELECT? =3
```

In the program is possible to select one of the following option:

```
OPTION 1    DATA
OPTION 2    START
OPTION 3    EXIT – GO BACK TO THE FIRST MASK OF THE MAIN MENU'
```

Confirm with E to go on with the selected option.

3.6.1 DATA

Before starting the sample is possible to do a check / modify of the programming data of the sample itself. The programming data remains in memory until the next modification.

```
ISOKINETIC SAMPLING  <S>SET  <E>CONT
1=SET TIME  2=SET VOLUME      =1
```

Set up the programming methods. Select the option 1 and it will be set the sampling time, otherwise, with the option 2, it will be set the volume to sample.

Press E to go on or press the SET button to change the proposed value.

```
ISOKINETIC SAMPLING  <S>SET  <E>CONT
SAMPLING TIME (2-9999)      min=  60
```

The mask with the sample length modification will be displayed if at the previous question the answer was the option 1, otherwise will be displayed the following mask:

```
ISOKINETIC SAMPLING  <S>SET  <E>CONT
LITERS TO SAMPLE (1-65000)  1= 100
```

Press E to go on or press the SET button to change the proposed value.

```
ISOKINETIC SAMPLING  <S>SET  <E>CONT
NORMALIZED TEMPERATURE (0-50)  C= 20
```

Set the normalization temperature.

Press E to go on or press the SET button to change the proposed value.

```
ISOKINETIC SAMPLING  <S>SET  <E>CONT
LITERS ASP. DERIVATE LINE (0-10)  L= 2
```

Set aside stream flow.

Press E to go on or press the SET button to change the proposed value.

Setting "0" means side stream not enable, and next mask is set Delay of start sampling.
Setting a flow, the next mask is:



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ISOKINETIC SAMPLING	<S>SET	<E>CONT
TEMP.GAS METER DERIV. LINE (0-50)C=	25	

Set temperature of the Dry gas meter of side stream line.
Press E to go on or press the SET button to change the proposed value.

ISOKINETIC SAMPLING	<S>SET	<E>CONT
SAMPLING DELAYED START (1-9999)min =	1	

Set delay needed before to start the sampling.
Press E to go on or press the SET button to change the proposed value, that it is always equal to zero.

ISOKINETIC SAMPLING	<S>SET	<E>CONT
MEASURE UNIT (0=mmH2O 1=Pa) =	1	

Set Unit Measure of static and differential pressure.
Press E to go on or press the SET button to change the proposed value.

NOZ	2	4	6	8	10	<E>CONT
	1.2	4.7	10.6	18.8	29.4	

At this point will be proposed a table containing a series of nozzle and their isokinetic flows.

This table will be compiled with the values calculated in real time using differential pressure data, static pressure and duct temperature retrieved from the ISOCHECK instrument.

Press E to go on.

ISOKINETIC SAMPLING	<S>SET	<E>CONT
NOZZLE DIAMETER	mm =	6

It will be proposed the diameter of the nozzle to use.
Press E to go on or press the SET button to change the proposed value.

NOZZLE DIAMETER = 6	<SET>CHANGE	<E> CONT
ISOKINETIC FLOW =	10.57	

Press E to go on or press the SET button to go back to the previous mask and change the nozzle diameter and show their isokinetic flows.

***	AUTOMATIC ISOKINETIC SAMPLING	***
1=DATA	2=START	3=EXIT
	SELECT?	=3

At the end the instrument will return to the "AUTOMATIC ISOKINETIC SAMPLING" program menu'.



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3.6.2 START

After all the pneumatic links are set up is possible to select this option to start the sampling.

The sampling will be done using the settings of the "DATA" section.

If it was selected a delay, the next mask appears:

DELAY START TIME min = 5
Sec. COUNT = 300 <C>STOP

At the end of count down the system start the sampling.

Press C to stop the count down and come back to main menu.

WARNING!

If the ISOCHECK does not measure any duct flow rate, the system will report that the CALCULATED isokinetic flow is less than 1l/min. After that the instrument will recalculate the FLOW until the intervention of the operator or the presence of a scope that determine a theoretical FLOW > 1 l/min (in this cases the system will start automatically)

At the start of the sampling the follow mask will appear.

*****	STACK SAMPLING	*****
***	WAIT! FLOW REGULATION	***

After a few seconds these masks will be sequentially displayed.

TIME	TEMP	VOLUME	ISO.FL	FLOW	F LD
02.2	28.6	00023	10.57	10.48	-0.8

... 1^ mask (displayed alternatively at the 2^) ...

DP	T.stack	STATIC P.	SPEED	NZ
5.0	0099	180	8.56	06

... 2^ mask (displayed alternatively at the 1^) ...

FLOW RATE Nm3/h	VACUUM mmhg	% ERROR
353660	180	-0.75

... 3^ mask ...

During the sampling will be displayed the following data:

1^ MASK:

TIME	Tempo (minutes) - This value increase while the sampling goes on
TEMP	Temperature (°C) at the counter - This value increase while the sampling goes on
VOLUME	Sampled Volume (litres) - This value increase while the sampling goes on



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ISO.FL	ISOKINETIC Flow (l/min) – This values are calculated in real time
FLOW	EFFECTIVE Flow (l/min), This value means the actual sampling flow of the LIFETEK. The system will follow the isokinetic condition correcting the sampling flow. In case of the side stream is enable, the isokinetic flow will be always regulated in order to maintain right flow as per formula: $FLOW + F LD$.
F LD	Side Stream Flow (l/m) based on the dry gas meter temperature of side stream LIFETEK

2^ MASK:

DP	Differential pressure (value acquired from the ISO CHECK)
T.stack	Duct temperature (value acquired from the ISO CHECK)
STATIC P.	Static duct pressure (value acquired from the ISO CHECK)
SPEED	Speed
NOZ	Selected Nozzle

3^ MASCHERA:

FLOW RATE	Q (Nm ³ /h) – Hour average of stack flow rate emission
VOID	Residual Vacuum (mmhg) of sampling line
ERROR Error (%)	Error (%) of actual flow (FLOW) in refer to of flow account (ISO.FL). In case of side stream enable, the % error is calculated as actual flow (FLOW + F LD) in refer to counted flow (ISO.FL).

During the sampling the key is active ← (C). Pressing this button is possible to stop earlier the sampling.

According to the norm, the instrument every 5 minutes display ythe following parameters:

```
mmH2O; t_cont; l= 80;
p_d=10.00; p_s= 0; t_c= 0; v=10.30
```

mmH2O = Unit of measurement of pressure
t_cont = Dry gas meter temperature
l = Sampled liter up to this time
p_d = Differential pressure
p_s = Static pressure
t_c = Duct temperature
v = Flow speed

At the end of the sampling will be displayed the following mask:

```
END SAMPLING
SAVING DATA
```



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At the end of the sample the pump will stop, the sampling result are stored and the device will automatically go back to the first mask of the main menu'. The stored data can be displayed, printed, downloaded to a PC or deleted using the data management menu' (see the relative paragraph).

EXIT - GO BACK TO THE FIRS MASK OF THE MAIN MENU'

Select this option to go back to the 1^ mask of the main menu'.

SELECT PROGRAM F1>> F2<< <E>CONT TEST (TEMPERATURA-LEAK-LOAD-FLOW)
--

1^ mask of the main menu'

3.7 ISOKITENITC SAMPLING WITH TSB

With the ISOKINETIC SAMPLING WITH TSB is possible to do AUTOMATIC ISOKINETIC samples.

To do the automated sample is necessary to have an automated ISOCHECK TSB processor.

- Do all the electric and pneumatic links of the sampler
- Do the link between the sampler and the processor using a serial cable.

On the LIFETEK sampler do as follow:

Use the F1 and F2 keys to scroll the main menu'. Select "ISOKINETIC SAMPLING WITH ISOCHECK" and press the E button to gain access to the program.

** TSB SAMPLING ** <C> RETURN
--

At this point it is possible to manage sample from the TSB.

The data related at the sampling will be set and stored on the TSB.

To exit the program and go back to the main menu' press the key ← (C button).

IMPORTANT:

AFTER 5 MINUTES WITHOUT ANY COMMUNICATION BETWEEN ISOCHECK TSB and LIFETEK, A MESSAGE ON DISPLAY APPEAR "COMMUNICATION ERROR".

THE MESSAGE DISAPPEAR WHEN THERE WILL BE THE FIRST SIGNAL COMMUNICATION



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3.8 MEMORIZED DATA MANAGMENT

Use the F1 and F2 keys to scroll the main menu'. Select "STORED DATA" and press the E button to gain access to the program.

In the program is possible to select one of the following option:

- OPTION 1 SHOW DATA FROM MEMORY
- OPTION 2 DATA DOWNLOAD
- OPTION 3 PRINT DATA FROM MEMORY
- OPTION 4 ERASE MEMORY
- OPTION 5 BACK TO MAIN MENU'

Use the F1 and F2 keys to scroll the **stored data menu'**. Press the E button to select the desiderated option.

3.8.1 SHOW DATA FROM MEMORY

Select this option to display the stored data.

```
SELECT SAMPLING
0=ENV 1=COND 2=ISO 3=EXIT      SELECT? =3
```

In this mask is possible to choose the sampling methods to display.

The sampling methods are:

- OPTION 0 ENVIRONMENT DUST SAMPLING
- OPTION 1 DUCT DUST SAMPLING
- OPTION 2 AUTOMATIC ISOKINETIC SAMPLING

The option 3 send you back to the main mask of the memorized data management.

The memory can store up to 50 FILE (in dynamic mode) for every type of sampling.
The total number of the file for one type of sampling can change depending on how many files of the other types are been stored.

Depending of the chosen sampling methods it will be displayed a mask like this:

```
SELECT SAMPLING
FILE SAMPLING AMBIENT 1
```

This mask is displayed as an example of the selection of the option 0.

The memory can save 50 FILES in dynamic way. The total number of saved file for each sampling changes in accord to dimension of saved file.

The next mask change depends on sampling selected:



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```
SELECT SAMPLING
FILE AMBIENT SAMPLING 1
```

Example mask 0 option

In this mask (showed for about 3 seconds) it's displayed the total number of the available file for this type of sampling. In the case that no files are available you will be warned that for the selected file type there aren't any stored data.

After 3 seconds will be displayed the following mask:

```
@4=01/01/06-12:00:00 ANVIROMENT SAMPL.
N.FILE=1      1=HOME 2=NEXT 3=EXIT E=CONT
```

The first line contains the identification data sampling, that is DATE and TIME start of the programmed sample.

The second line contains the option available for the selection/display of the data.

Press 2 to select the next sampling (if available in memory). The first line will be updated with the identification data of the next sampling available in memory and the counter file (N. FILE) will be increased.

After the file to review is selected, press the E button to scroll on the 1st line of the display all the related data of the sampling itself.

If during the data visualization you want to go back to the beginning of the file, is necessary to press the button 1.

At the end of the visualization press the button 3 to go back to the mask that allow to select the sampling methods.



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3.8.2 DATA DOWNLOAD

Select this option to download the stored data to a PC

To do the **download and the data management** is possible to use the Software LIFETEK XS 1.0 (optional).

To **download only the data** is possible to use the Hyper Terminal of Windows or similar software.

The first step to do is the setting of the serial port of the PC.

The port must be configured as follow:

Bit for seconds:	9600
Data Bit:	8
Parity:	None
Stop bit:	1
Flow check:	None

After that link the sampler to the serial port RS232 of the PC.

Activate the program on the PC as it remains in waiting mode for the data from the sampler.

On the sampler ...

DOWNLOAD DATA		
OPEN PC SOFTWARE	S=EXIT	E=CONT

Press the E button to start the download.

Press the SET button to exit and go back to the main mask of the memorized data management program.

At the end the instrument will automatically go back to the first mask of the memorized data management program.

At the next page will be reported an example of a downloaded file on a PC.



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@12 -09/03/2007-10:41:07- P.ISOCINETICO	Date/hour start program
SAMPLING TIME min = 10	DATA SET
SIZE STACK CIRCULAR	
STACK DIAMETER cm = 100	
PITOT COSTANT =0.83	
DENSITY Kg/m3 =1.293	
MOISTURE % =0.00	
BAROMETRIC hpa =1013	
TEMPERATURA NORM C.=20	
TEMPERATURA STACK C.= 100	
NOZZLE DIAMETER mm = 4	
@14 -10:41:23 DATI P.- t_cont=24.7	PARTIAL DATA
mmH2O; t_cont=27.0;l= 30	mmH2O =UNIT OF MEASUREMENT OF
p_d=10.00; p_s= -1; t_c= 50; v=11.12;	PRESSURE
@14 -15:14:12 DATI P.- t_cont=27.1; l= 30	T_cont = dgm TEMPERATURE
p_d=10.00; p_s= -1; t_c= 50; v=11.12@14 -	L = SAMPLED LITERS
	P_d. = DIFFERENTIAL PRESSURE
	P_s. = STATIC PRESSURE
	T_c = STACK TEMPERATURE
	v = SPEED
@13 -09/03/2007-10:51:07- END DATA	FINAL REPORT DATA
VOLUME ASPIRATO LIFETEK l = 233.0	
DURATA PRELIEVO min = 15.1	
FLUSSO MEDIO LIFETEK l/min = 15.48	
TEMP. MEDIA CONTATORE C. = 27.05	
VOLUME NORM. LIFETEK l = 212.0	
TEMP. MEDIA CONDOTTO C. = 50	
PRESS.STATICA ASSOLUTA hPa = 1012.9	
VELOCITA MEDIA m/sec = 11.12	
PORTATA MEDIA m3/h = 31445	
PORTATA MEDIA NORM. Nm3/h = 26572	
VOLUME LD (T.CONT LD) l = 30.1	
VOLUME LD(T.MEDIA CONT.) l = 33.1	
VOLUME TOT.(L.D.+LIFTEK) l = 266.0	
VOLUME LD (T. NORM.) l = 30.1	
VOLUME TOT. (T. NORM.) l = 242.1	
GRADO ISOCINETISMO % = 100.7	
SAMPLING OUTCOME : OK	
end file	



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3.8.3 PRINT STORED DATA

Select this option to print the stored data.

```
SELECT SAMPLING
0=ENV 1=COND 2=ISO 3=EXIT    SELECT? =3
```

In this mask is possible to choose the sampling methods to print.

The sampling methods are:

```
OPTION 0    ENVIRONMENT DUST SAMPLING
OPTION 1    DUCT DUST SAMPLING
OPTION 2    AUTOMATIC ISOKINETIC SAMPLING
```

Select the option 3 to go back to the main mask of the memorized data management program.

The option 3 means to come back at main menu.

Depending of the chosen sampling methods will be displayed a mask like this:

```
SELECT SAMPLING
FILE SAMPLING 1
```

This mask is displayed as an example of the selection of the option 0

In this mask (displayed for about 3 seconds) will be showed the total number of available file for this sampling methods. In case that no files exist, the user will be warned that for the chosen file type no stored data are available.

After 3 seconds will be displayed the following mask:

```
@4=01/01/06-12:00:00 PREL. AMBIENTE
N.FILE=1          1=STAMPA 2=NEXT 3=EXIT
```

The first line contains the identification data sampling, that is DATE and TIME start of the programmed sample.

The second line contains the option available for the selection/display of the data.

Press 2 to select the next sampling (if available in memory). The first line will be updated with the identification data of the next sampling available in memory and the file counter (N. FILE) will be increased.

After the file to print is selected press the button 1 to start printing.

At the end of the printing press the button 3 to go back to the mask that allow to select the sampling methods.



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Print example:

```
@4-01/01/06-15:48:55          PREL.
AMBIENTE
FLUSSO ASPIRAZIONE l/min = 10.00
FLUSSO STOP PREL.  l/min = 0.10
LITRI PRELIEVO      litri = 100
RITARDO PARTENZA   min = 0
TEMPERATURA NORM.  c. = 20
@8-01/01/06-15:57:39 DATI FINALI
LITRI ASPIRATI      litri = 83.5
DURATA PRELIEVO     min. = 8.7
FLUSSO MEDIO        l/min. = 9.58
TEMPERATURA MEDIA   c. = 28.53
LITRI NORMALIZZATI  nl/min.= 81.1
PRELIEVO FERMATO DALL'OPERATORE
```

3.8.4 TOTAL ERASE OF THE MEMORY

Select this option to totally erase the stored data. With this option **THE SETUP SETTINGS WILL NOT BE DELETED.**

```
**GESTIONE MEMORIA**  F1>> F2<< <E>CONT
PASSWORD ? = 1
```

Press the SET button to insert the password that is **1234**.

The password can't be changed, but will be request to make INTENTIONAL the data erase.

Press E to definitely erase the data.

At the end of the erasing memory process will be displayed the availability of the memory itself equal to 100% and after that the instrument will automatically go back to the memory menu'.

GO BACK TO THE MAIN MENU'

Select this option to go back to the 1^ mask of the memory menu'.

```
**SCELTA PROGRAMMA**  F1>> F2<< <E>CONT
TEST (TEMPERATURA-TENUTA-CARICO-FLUSSO)
```

1^ mask of the main menu'



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3.9 MISSING MAIN SUPPLY FUNCTION

The sampling is provided with a function called MISSING MAIN SUPPLY.

This function allows the sampler to stay in standby for about 4 hours in the case the main supply stop.

If during a sampling the main supply will interrupt will occur the following conditions:

- the pump stop
- after about 15/30 seconds from the main supply failure on the display will be showed the message "MISSING MAIN SUPPLY" and the sampler will be in standby

When the main power is restored the pump start again and the sampling will continue from where it stopped.

The display will be automatically "refreshed" and it will go back to the conditions before the main power stopped.

The signalling with the indication of the stopping time and the restore time will be stored together with the final data.



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4 WARNINGS

Be sure that the power supply cable is well connect in a correct socket of the right voltage and not superior. Connect the sampler on the electrical grid of 230Vac – 50Hz with grounding.

Only LIFETEK 33 and 55 – Use only battery with the correct voltage wrote on the back panel of the instrument (12Vcc and 24Vcc). **Be very careful about the connection cable polarity.**

(RED= POSITIVE POLE / BLACK= NEGATIVE POLE)

Do not use the sampler if the main supply cable or the socket are damaged or the power plug is not completely connected in the socket.

Do not block the side air vent to prevent the pump overheat.

Do not disassemble or modify any parts of the sampler. Internal inspection, modification or reparations must be done only to qualified technical staff.

Do not disassemble, change or alter the internal batteries (if present).

Do not cut, damage or alter the main supply cable. That operation can cause a short-circuit.

The sampler must avoid any contact with water or other fluid substance. Contact the technical assistance.

Do not immerse the sampler in water or other type of fluid substance.

Do not touch the main supply cable or the sampler with wet hands. That could cause electrical shock.

The glass impinger's liquid bubble or the condense must not be inspired and enter in the pump.

If it happens disconnect the sampling line and do a sample with the pump at the maximum for a few minutes. Contact the technical assistance.

Don't do TEST – LOAD TABLE with in line glass impingers.

Avoid to use, put or store the sampler in place exposed to high sunlight or high temperature. The direct solar exposition and heat could cause the leak of liquids from the internal battery (if present) or cause the overheating or the explosion causing fire, burns or other injuries.



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Stop immediately the use of the sampler if it falls or it's damaged or if it emits smoke or noxious fumes.

Shut it down and remove the internal battery or disconnect the main supply.

To remove the internal battery (if any) unscrew the lives in the cover and open the sampler.

Contact the technical assistance.

Do not store the sampler in humid or dust place.

Disconnect the main supply from the sampler and from the socket after the use and when the sampler is no powered on to avoid fire or other dangers.

Avoid to leave the sampler unused for long time. It's recommended to use it, vacuum, once a month for at least 15 minutes. After that do a full charging cycle and store it.



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5 TROUBLESHOOTING

5.1 ANOMALIES

It follows a table with the most likely anomalies, their probable causes and their possible verify operation to solve the problem.

In the case it will not be possible to solve the problem contact the technical assistance.

ANOMALIE	CAUSE	SOLUTION
DOESN'T POWER UP	The sampler is off	Turn it on using the ON/OFF switch
	The main supply cable or the external battery cable aren't connected	Connect the main supply cable and use the ON/OFF switch otherwise connect the external batteries at 24Vcc
	The fuse situated in the sampling line is blown.	Substitute the fuse with the spare one. If the problem persists, contact the technical assistance.
THE PUMP DOESN'T START	The pump is broken	Contact the technical assistance
WHEN THE MAIN POWER IS MISSING THE FUNCTION "WAITING NO POWER" DOESN'T START	The internal batteries wasn't used for a long time	Use the sampler with the main supply to recharge the internal batteries.
	The internal batteries have lost effectiveness.	Contact the technical assistance
THE PUMP DOESN'T REACH THE DESIDERED SCOPE	The pump is dirty because some glass impinger's liquid or chimney condense is entered it.	Contact the technical assistance
	The type of filter used creates a too high load loss.	Use a filter with a major diameter and/or major porosity
THE PUMP DOESN'T KEEP THE SELECTED SCOPE	The filter is clogged	End the sampling
	The tube that connect the sensor to the sampler is blocked.	Remove the obstruction



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6 TECHINCAL SPECIFICATION

LIFETEK 33 XP - R		
Pump	Membrane	High pump head > 400 mmHg (30l/min)
Flow rate	Range	20 cc/min – 25 l/min
Dry gas meter		Error max $\pm$ 2%
Power supply		230 Vac – 50 Hz
Operative temperature		-10 °C ÷ +45 °C 95%UR.
Flow digital control (flom meter not needed)		
Automatic system for pressure drop compensation		
Vacuum digital meter		
Setting of the flow by keypad.		
Alarm Management		
System in compliance with UNI EN ISO 16911-1 e UNI EN 13284-1		
Environmental sampling at constant flow (es pm10-pm2.5)		
Isokinetic Sampling semi-automatic with manual program of the parameters		
Isokinetic Sampling automatic with connection of ISOCHECK, with automatic regulation of the flow measuring stack parameters		
Serial 232 interface to download data by PC		
Serial Interface with ISOCHECK for isokinetic sampling		
Internal Memory to save sampling data		
Automatic restart after Power failure internal battery) Power failure events recorded by alarm		
Power supply by external battery of 12Vcc (optional).		
Dimension		240 x 260 x 285 mm
Weight		7 Kg



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LIFETEK 55 XP - R		
Pump	Membrane	High pump head > 400 mmHg (60l/min)
Flow rate	Range	1 l/min – 50 l/min
Dry gas meter		Error max $\pm$ 2%
Power supply		230 Vac – 50 Hz
Operative temperature		-10 °C ÷ +45 °C 95%UR.
Flow digital control (flom meter not needed)		
Automatic system for pressure drop compensation		
Vacuum digital meter		
Setting of the flow by keypad.		
Alarm Management		
System in compliance with UNI EN ISO 16911-1 e UNI EN 13284-1		
Environmental sampling at constant flow (ex pm10-pm2.5)		
Isokinetic Sampling semi-automatic with manual program of the parameters		
Isokinetic Sampling automatic with connection of ISOCHECK, with automatic regulation of the flow measuring stack parameters		
Serial 232 interface to download data by PC		
Serial Interface with ISOCHECK for isokinetic sampling		
Internal Memory to save sampling data		
Automatic restart after Power internal battery failure. Power failure events recorded by alarm		
Power supply by external battery of 24Vcc (optional).		
Dimension		280 x 300 x 270 mm
Weight		11 Kg



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LIFETEK 100 XP - R		
Pump	Membrane	High pump head > 400 mmHg (100l/min)
Flow rate	Range	5 l/min – 70 l/min
Dry gas meter		Error max $\pm 2\%$
Power supply		230 Vac – 50 Hz
Operative temperature		-10 °C ÷ +45 °C 95%UR.
Flow digital control (flow meter not needed)		
Automatic system for pressure drop compensation		
Vacuum digital meter		
Setting of the flow by keypad.		
Alarm Management		
System in compliance with UNI EN ISO 16911-1 e UNI EN 13284-1		
Environmental sampling at constant flow (ex pm10-pm2.5)		
Isokinetic Sampling semi-automatic with manual program of the parameters		
Isokinetic Sampling automatic with connection of ISOCHECK, with automatic regulation of the flow measuring stack parameters		
Serial 232 interface to download data by PC		
Serial Interface with ISOCHECK for isokinetic sampling		
Internal Memory to save sampling data		
Automatic restart after Power internal battery failure. Power failure events recorded by alarm		
Power supply by external battery of 24Vcc (optional).		
Dimension		365 x 375 x 345 mm
Weight		20 Kg

SENSORI (per tutte le versioni)		
Dry gas meter temperature	Range	0 ÷ 50 °C
	Resolution	0,1 °C
	Precision	$\pm 2^\circ\text{C}$
	Linearity	$\pm 2^\circ\text{C}$ software
Vacuum	Range	0 ÷ 760 mmHg
	Resolution	1 mmHg
	Precision	± 5 mmHg



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7 DECLARATION OF CONFORMITY

Dichiarazione di conformità UE

EU Declaration of Conformity

Nome del fabbricante / Company name:	Mega System srl
Indirizzo postale / Postal address:	Via don Fracassi 41/41
CAP e Città / Postcode and City:	20010 Bareggio, Milan, Italy
Telephone / Telephone:	+39 02 90361622
Indirizzo Posta elettronica / E-Mail address:	info@megasystemsrl.com

dichiara che la presente dichiarazione di conformità UE è rilasciata sotto la responsabilità esclusiva del fabbricante.

declare that the Declaration of conformity is issued under our sole responsibility:

Oggetto della dichiarazione / Object of the declaration:

Series / Serie:	LIFETEK 33-55-100
Product Type / Tipo prodotto:	CAMPIONATORI AMBIENTALI / ENVIRONMENTAL SAMPLER

L'oggetto della dichiarazione di cui sopra è conforme alla pertinente normativa di armonizzazione dell'Unione /

The object of the declaration described above is in conformity with the relevant Union harmonisation legislation:

- **Direttiva 2006-42-CE del 17-06-2006 / Directive 2006/42/CE of 17-06-2006**
- **Direttiva 2014-30-UE del 26-02-2014 / Directive 2014/30/UE of 26-02-2014**
- **Direttiva 2014-35-UE del 26-02-2014 / Directive 2014/30/UE of 26-02-2014**

Bareggio, 14/09/2016

Mega System srl
Amm. Unico / sole administrator
G.Colnago



Pollution Control Systems

LIFETEK 33/55/100

GENERAL CONDITIONS FOR WARRANTY

Dear Customer,

Thanking you for purchasing a Mega System instrument we would like to remember that the warranty is valid for twelve months starting from the date of the transport document.

The warranty DO NOT include damages caused by misuse of the instrument, as well as natural events (fire, collapse, flooding etc.), falling, acts of violence, tampering and connection to electrical systems that don't comply.

Are also not include any parts that you consume under normal use of instrument itself.

Any repairs under warranty must be made at our laboratory at the following address

Via Don Fracassi 41 – 20010 Bareggio (Milano) - ITALY

At the receiving of the material our technicians will provide the restore/substitution of the parts that will be judged, at their sole discretion, defective. In case of, instead, the registered anomaly is not covered by the warranty conditions, will be invoiced at list price the labour and any spare parts, independently from the date of the transport document, prior client approval by a written quotation.

THE SHIPMENT COSTS ARE CHARGED TO THE CUSTOMERS EXCEPT DIFFERENT AGREEMENT

We suggest to contact our technical assistance, to the following contact details, before shipping the instrument.

Best Regards

Mega System Srl
Technical Assistance Service
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Pollution Control Systems

LIFETEK 33/55/100

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DOCUMENT REVISION	DATE	DESCRIPTION
00	09/2007	First edition
01	05/2016	General revision
02	09/2016	EU declaration

SW REVISION	DATE	DESCRIPTION
SW 2.5	09/2012	General revision – manual validity LIFETEK 33/55/100 (XP-R)
SW 2.6	03/2013	Upgrade Leak test
SW 2.7	07/2013	Show QVN at place of QV in isokinetic flow
SW 2.8	10/2013	UNI 16911-1
SW 2.9	11/2013	Leak test modification
SW 3.6	07/2016	Remote link data